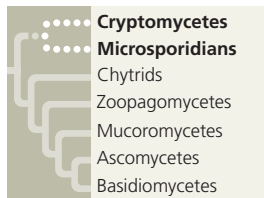


Cryptomycetes and Microsporidians



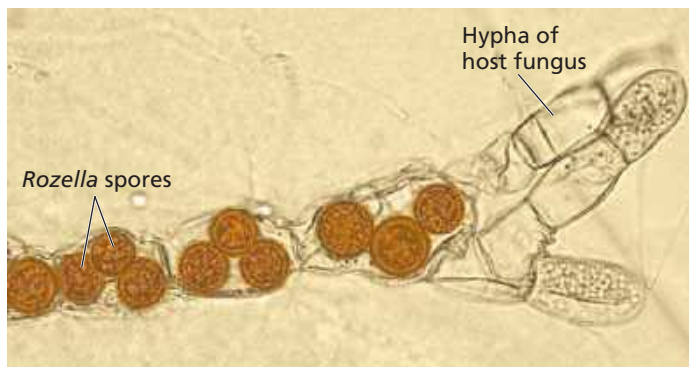
Genomic studies indicate that **cryptomycetes** (fungi in the phylum Cryptomycota) and **microsporidians** (fungi in the phylum Microsporidia) form a sister group and are a basal fungal

lineage (see Figure 31.10). While most molecular comparisons support the placement of cryptomycetes and microsporidians at the base of the fungal tree, more data are needed to help resolve this phylogeny.

Cryptomycetes

Although only 30 species have been identified to date, genetic data suggest that the cryptomycetes are a large and diverse group. DNA sequences from members of this group have been found in marine and freshwater communities, as well as soils. Cryptomycetes also have been found in aerobic and anaerobic environments, and in geographical locations across the globe. Like the species shown in [Figure 31.11](#), *Rozella allomycis*, many of the cryptomycetes identified to date are parasites of protists and other fungi.

▼ **Figure 31.11** The cryptomycete *Rozella allomycis* parasitizing another fungus.



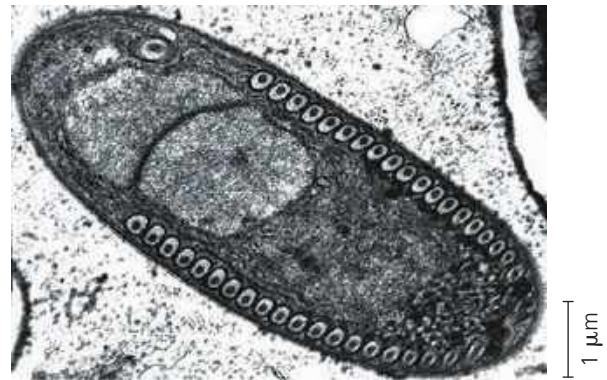
Cryptomycetes are unicellular and have flagellated spores. Cryptomycetes also can synthesize a chitin-rich cell wall, a key structural feature of the fungi (see Concept 31.1).

Microsporidians

The 1,300 species of microsporidians are unicellular parasites of protists and animals, including humans ([Figure 31.12](#)). Infections in humans can cause reduced longevity and weight loss. The microsporidian *Nosema ceranae* is a parasite of honeybees and may contribute to *Colony Collapse Disorder*, a devastating outbreak that has led to the loss of honeybee colonies throughout the world.

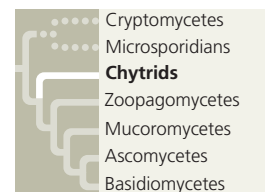
Like all fungi, microsporidians can synthesize a chitin-rich cell wall. Other aspects of their biology are unusual. For example, microsporidians have highly reduced mitochondria and

▼ **Figure 31.12** A spore of the microsporidian *Fibrillanosema crangonycis*. (TEM)



small genomes, with only 2,000 genes in some species. The genome of one microsporidian, *Encephalitozoon intestinalis*, has just 2.3 Mb of DNA—the smallest genome of any eukaryote sequenced to date. Unlike other basal fungi, microsporidians lack flagellated spores; instead, they produce unique spores that infect host cells via a harpoon-like organelle.

Chytrids



The fungi classified in phylum Chytridiomycota, called **chytrids**, are ubiquitous in lakes and soil; recent metagenomic studies have uncovered new clades of chytrids in hydrothermal vent and other

marine communities. Some of the approximately 1,000 chytrid species are decomposers, while others are parasites of protists, other fungi, plants, or animals; as we'll see later in the chapter, two chytrid parasites have contributed to the global decline of amphibian populations. Still other chytrids are important mutualists. For example, anaerobic chytrids that live in the digestive tracts of sheep and cattle help to break down plant matter, thereby contributing significantly to the animal's growth.

Nearly all chytrids have flagellated spores, called **zoospores** ([Figure 31.13](#)). Like other fungi, chytrids have cell walls made of chitin, and they also share certain key enzymes and metabolic pathways with other fungal groups. Some chytrids form colonies with hyphae, while others exist as single spherical cells.

➔ **Mastering Biology Video: Phlyctochytrium Zoospore Release**

► **Figure 31.13** Flagellated chytrid zoospore. (TEM)

